RTCA, Inc.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 25

Electrostatic Discharge (ESD)

Important Notice

Information pertinent to this test procedure is contained in Sections 1, 2, 3 and 20. Further, Appendix A is applicable for identifying the environmental tests performed.

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25 Electrostatic Discharge (ESD)

25.1 Scope

The Electrostatic Discharge test relates to airborne equipment which may be involved in static electricity discharges from human contact. Some factors contributing to an ESD event may be: low relative humidity, temperature, use of low conductivity (artificial fiber) carpets, vinyl seats and plastic structures which may exist in all locations within an aircraft. This test is applicable for all equipment and surfaces which are accessible during normal operation and/or maintenance of the aircraft. This test is not applicable to connector pins.

25.2 Purpose of the Test

The electrostatic discharge test is designed to determine the immunity or the ability of equipment to perform its intended function without permanent degradation of performance as a result of an air discharged electrostatic pulse.

25.3 Test Description

The immunity to electrostatic discharge shall be determined by the ability of the equipment under test (EUT) to withstand a series of electrostatic discharge pulses at a test level of 15,000 volts, directed at specific human contact locations on the EUT. The quantity of pulses shall be ten (10) in each of the selected locations and in both positive and negative voltage polarities (10 positive and 10 negative). The test configuration is depicted in Figure 25-1.

25.4 Equipment Categories

Category A - Electronic equipment that is installed or operated during normal operation and/or maintenance of the aircraft.

25.5 Test Procedure

With the equipment powered and operated in the required mode the electrostatic discharge test will be performed under the following conditions:

25.5.1 **Test Configuration** - The EUT shall be set up as shown in Section 20.3a (General Test Requirements). Connect and orient the equipment as specified by the applicable installation and interface control drawings or diagrams. Care must be taken in routing and grounding of the cabling associated with the ESD generator to minimize the potential for secondary effect of the radiated field from the cabling. This test is intended to test the primary effect which is the discharge from the ESD generator to the enclosure of the equipment under test. This includes the normal method of mounting, bonding, and grounding of the equipment.

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25.5.2 **ESD Generator** - The ESD generator shall have a general schematic as shown in Figure 25-2, with a discharge resistor of 330 ohms (±20%) and an energy storage capacitor of 150 pf (±20%), and shall be capable of generating a pulse of 15,000 volts, in both positive and negative polarities. The ESD generator shall also have an air discharge tip as shown in Figure 25-3. The ESD generator shall be calibrated to produce a positive and negative 15,000 volt (+10%, -0%) peak output pulse. The generator setting required to produce this output shall be recorded.

25.5.3 **EUT Test Modes** - EUT test modes shall include software (if applicable) chosen to exercise all normal modes of operation of the EUT.

Note: If monitoring equipment is required, it should be decoupled in order to reduce the possibility of erroneous failure indications.

25.5.4 **Selection of Test Points** - The test points to be considered shall include the following locations as applicable: any conductive or non-conductive point in the control or keyboard areas and any other point of human contact, such as switches, knobs, buttons, indicators, LEDs, seams, slots, grilles, connector shells and other operator accessible areas.

25.5.5 **Pulse Application** - The ESD discharges shall be applied only to those points and surfaces of the EUT which are accessible to personnel during normal operation (including as installed on aircraft maintenance).

With the ESD generator set at the value recorded during calibration, the ESD generator shall be held perpendicular to the surface to which the discharge is to be applied. The discharge return cable of the generator shall be grounded to the ground plane and kept at a distance of at least 0.2 meters from the EUT and its cabling.

25.5.6 **Test Technique** - Move the tip of the ESD generator toward the test point at the same speed a human hand would reach to touch an object (approximately 0.3 meters/second) until the generator discharges or until contact is made with the test point. After each discharge or attempted discharge, the ESD generator (discharge electrode) shall be removed from the test point. The generator is then recharged for the next single discharge. This procedure shall be repeated until the 10 discharges or attempted discharges in each polarity and each test point are completed. In the event of no discharge for a selected test point, attempt to locate a discharge point in the area around the test point, e.g., attempt to discharge to multiple points along a section of seam or slot that is identified as a single test point or around all sides of a button that is identified as a single test point.

25.6 Evaluation of the Test Results

Following application of the pulses, DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS, unless specified otherwise.

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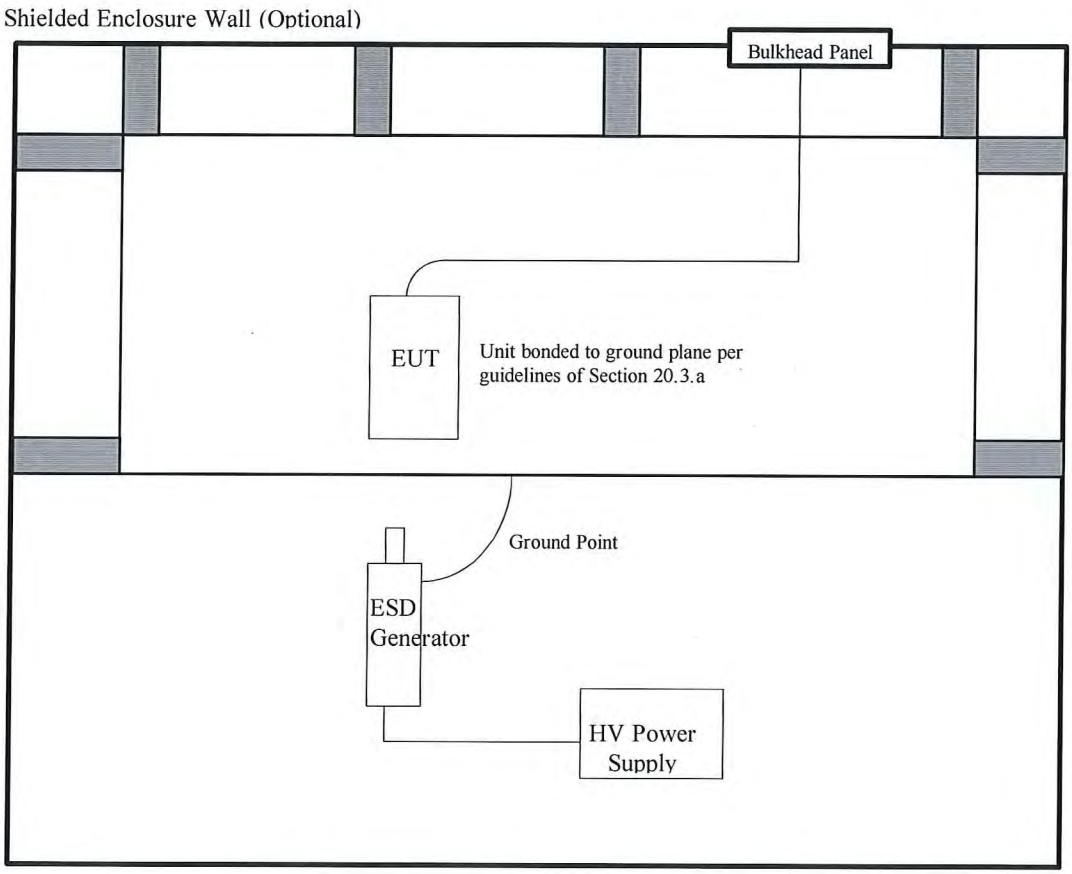

Figure 25-1 **Electrostatic Discharge Typical Test Setup**

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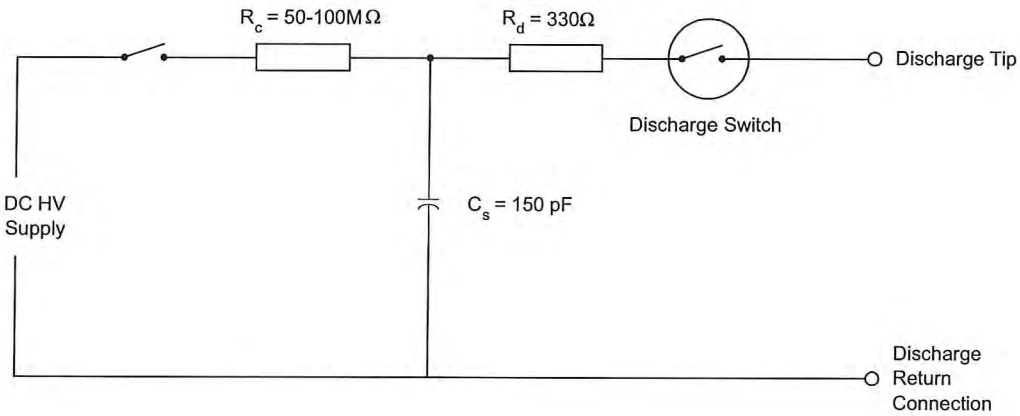

Figure 25-2 Simplified Diagram of the ESD Generator

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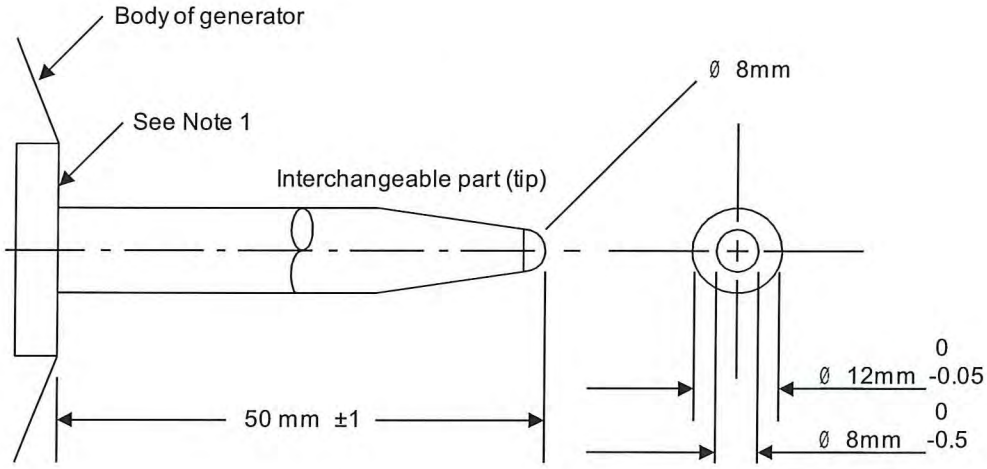

Note 1 - The discharge switch (e.g., vacuum relay) shall be mounted as close as possible to the tip of the discharge electrode.

Figure 25-3 Discharge Electrode of the ESD Generator

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